

# Mixed-Precision Randomized Sketching for Scaled Dot-Product Attention

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**Abstract.** We propose a mathematically focused study of scaled dot-product attention under two simultaneous approximations: randomized feature sketching and mixed-precision arithmetic. The exact logit operator  $d^{-1/2}QK^T$  is replaced by the sketched operator  $d^{-1/2}QSS^TK^T$ , where  $S$  is a random sketching matrix of smaller column dimension. The report isolates three levels of perturbation: the deterministic error in the logits, the induced perturbation of the rowwise softmax map, and the additional finite-precision error introduced when the sketched computation is carried out in low precision. The main analytical result is a clean error decomposition showing that the total output error is controlled by the sum of a sketching term and a rounding term. The accompanying experiments are designed as a validation of this decomposition rather than as a large-scale engineering benchmark.

Keywords: mixed-precision arithmetic, randomized numerical linear algebra, attention, transformers, sketching

## 1 Proposal

The purpose of this report is to study one analytically transparent fragment of the transformer architecture [1]: scaled dot-product attention. The report is intentionally not about full transformer training. Instead, it isolates a single nonlinear operator and examines how two approximations interact:

- (1) a randomized finite-rank compression motivated by randomized numerical linear algebra [2] [3];
- (2) a mixed-precision realization motivated by modern low-precision training practice [4].

This focus keeps the project aligned with the mathematical language of the lecture notes: dimension reduction, kernel-style matrix constructions, and operator perturbation. It also places the work in the neighborhood of efficient-attention models such as Linformer, Nyströmformer, and Performer [5] [6] [7], while remaining much narrower in scope.

The guiding question is the following.

**Idea 1** (Research question). When the attention logits are first compressed by a randomized sketch, can the subsequent computation be carried out in mixed precision with ↴

only a lower-order loss of accuracy? ┐

The intended contribution of the report is modest but precise: a deterministic error decomposition, together with numerical experiments that test whether the finite-precision term is indeed subordinate to the sketching term in relevant parameter regimes.

## 2 Operator-Theoretic Setup

Let  $\mathcal{H}_d := \mathbb{R}^d$  and  $\mathcal{H}_n := \mathbb{R}^n$ , both endowed with their standard Euclidean inner products. We identify matrices in  $\mathbb{R}^{n \times d}$  with Hilbert–Schmidt operators

$$Q, K, V : \mathcal{H}_d \rightarrow \mathcal{H}_n.$$

Their Hilbert–Schmidt norm is denoted by  $\|\cdot\|_{\text{HS}}$ , while  $\|\cdot\|_2$  denotes the spectral norm on linear operators.

The exact logit operator is

$$\mathcal{L}(Q, K) := d^{-1/2} Q K^* \in \mathcal{L}(\mathcal{H}_n),$$

and the rowwise softmax map is denoted by

$$\Sigma : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}.$$

Accordingly, the attention output is the nonlinear operator

$$\mathcal{A}(Q, K, V) := \Sigma(\mathcal{L}(Q, K))V.$$

Now let  $S \in \mathbb{R}^{d \times s}$  with  $s < d$ . The associated rank- $s$  positive semidefinite operator on  $\mathcal{H}_d$  is

$$P_S := S S^*.$$

The sketched logit operator is then

$$\tilde{\mathcal{L}}_S(Q, K) := d^{-1/2} Q P_S K^*,$$

and the corresponding sketched attention output is

$$\tilde{\mathcal{A}}_S(Q, K, V) := \Sigma(\tilde{\mathcal{L}}_S(Q, K))V.$$

Thus the basic approximation step is the replacement of the identity on  $\mathcal{H}_d$  by the finite-rank operator  $P_S$ .

## 3 Analytical Core

The analysis begins with a stability property of the rowwise softmax map.

**Proposition 2** (Rowwise softmax is Hilbert–Schmidt Lipschitz). Let  $L, M \in \mathbb{R}^{n \times n}$ . Then

$$\|\Sigma(L) - \Sigma(M)\|_{\text{HS}} \leq \|L - M\|_{\text{HS}}.$$

Consequently, for every  $V \in \mathbb{R}^{n \times d}$ ,

$$\|\Sigma(L)V - \Sigma(M)V\|_{\text{HS}} \leq \|L - M\|_{\text{HS}} \|V\|_2.$$

**Proof.** Let  $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$  be the ordinary softmax map on vectors. Its Jacobian at  $x \in \mathbb{R}^n$  is

$$D\sigma(x) = \text{Diag}(p) - pp^T, \quad p := \sigma(x).$$

For any  $z \in \mathbb{R}^n$  one has

$$z^T D\sigma(x) z = \sum_{i=1}^n p_i z_i^2 - \left( \sum_{i=1}^n p_i z_i \right)^2 \leq \sum_{i=1}^n p_i z_i^2 \leq \|z\|_2^2.$$

Hence  $\|D\sigma(x)\|_2 \leq 1$  for all  $x$ , and the mean value theorem yields

$$\|\sigma(x) - \sigma(y)\|_2 \leq \|x - y\|_2.$$

Applying this estimate row by row gives

$$\|\Sigma(L) - \Sigma(M)\|_{\text{HS}}^2 = \sum_{i=1}^n \|\sigma(\ell_i) - \sigma(m_i)\|_2^2 \leq \sum_{i=1}^n \|\ell_i - m_i\|_2^2 = \|L - M\|_{\text{HS}}^2,$$

where  $\ell_i$  and  $m_i$  are the rows of  $L$  and  $M$ . The second assertion follows from the submultiplicativity estimate

$$\|XV\|_{\text{HS}} \leq \|X\|_{\text{HS}} \|V\|_2.$$

□

The next proposition isolates the deterministic sketching error.

**Proposition 3** (Deterministic sketching bound). Let  $U \subseteq \mathcal{H}_d$  be a subspace containing all row vectors of  $Q$  and  $K$ . Assume that  $P_S$  is an  $\epsilon$ -approximate identity on  $U$ , in the sense that

$$\|(I - P_S)u\|_2 \leq \epsilon \|u\|_2 \quad \text{for all } u \in U.$$

Then

$$\|\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2.$$

Consequently,

$$\|\mathcal{A}(Q, K, V) - \tilde{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2 \|V\|_2.$$

**Proof.** Since

$$\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K) = d^{-1/2} Q(I - P_S) K^*,$$

we obtain

$$\|\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \leq \frac{1}{\sqrt{d}} \|Q(I - P_S)\|_{\text{HS}} \|K^*\|_2.$$

□

↳ Every row of  $Q$  lies in  $U$ , so the approximate-identity assumption implies

$$\|Q(I - P_S)\|_{\text{HS}} \leq \epsilon \|Q\|_{\text{HS}}.$$

Using  $\|K^*\|_2 = \|K\|_2$  proves the first estimate. The second follows from the previous proposition.  $\square$

This reduces the sketching analysis to a single geometric requirement on the projector  $P_S$ . In the randomized setting, the role of randNLA is to choose  $S$  so that this requirement holds with high probability on a subspace that captures the geometry of the rows of  $Q$  and  $K$ .

The final step is to separate sketching error from finite-precision error.

**Proposition 4** (Error splitting for mixed-precision attention). Let  $\hat{\mathcal{L}}_S(Q, K)$  denote the logit matrix produced by a finite-precision implementation of the sketched computation, and define

$$\hat{\mathcal{A}}_S(Q, K, V) := \Sigma(\hat{\mathcal{L}}_S(Q, K))V.$$

Then

$$\|\mathcal{A}(Q, K, V) - \hat{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \left( \|\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K)\|_{\text{HS}} + \|\tilde{\mathcal{L}}_S(Q, K) - \hat{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \right) \|V\|_2.$$

**Proof.** Apply the previous Lipschitz estimate with

$$L = \mathcal{L}(Q, K), \quad M = \hat{\mathcal{L}}_S(Q, K),$$

and then use the triangle inequality

$$\|L - M\|_{\text{HS}} \leq \|L - \tilde{\mathcal{L}}_S(Q, K)\|_{\text{HS}} + \|\tilde{\mathcal{L}}_S(Q, K) - M\|_{\text{HS}}.$$

$\square$

**Remark 5** (Interpretation). The first term in the bound is a purely sketching term; the second is a purely numerical term. In the intended mixed-precision regime, the numerical term should be lower order. This is the precise statement that the experiments are meant to test.  $\lrcorner$

## 4 Proposed Experiments

The numerical part of the project is not meant to be a large benchmark. Its role is to probe the analytical decomposition above. Accordingly, the experiments should compare three levels of computation:

- (1) exact unsketched attention in `float64`;
- (2) exact sketched attention in `float64`;
- (3) sketched attention in `float16` and mixed precision.

For each configuration we measure:

- the Hilbert–Schmidt error of the logits,
- the rowwise simplex error after softmax,
- the Hilbert–Schmidt error of the final attention output,
- the runtime.

The theoretical narrative suggests the following validation strategy. First, compare the exact unsketched and exact sketched logits; this isolates the sketching term. Second, compare the exact sketched and low-precision sketched logits; this isolates the numerical term. Finally, compare the resulting attention outputs. The main hypothesis is that once the sketch dimension  $s$  is moderate, the rounding term will be substantially smaller than the sketching term under mixed-precision accumulation, even if naive low precision is noticeably less stable.

## 5 Outlook

The present report deliberately studies only the feature-sketch model  $QK^T \approx QSS^TK^T$ . This is sufficient for a mathematically clean first analysis. If time permits, the natural extensions are Nyström-type approximations and random-feature approximations of the softmax kernel, which would connect the project more directly with the efficient-attention literature.